

STAT525 HOMEWORK #9

1. KNNL Problem 19.5
2. KNNL Problem 19.14
3. KNNL Problem 19.15
4. KNNL Problem 19.28
5. A clay tile company was interested in studying the effects of oven and cooling temperature on the strength of their tiles. The company's five ovens, used to bake the tiles, and four cooling temperatures ($^{\circ}\text{C}$) were considered. The data are shown below.

Cooling Temp	Oven					Mean
	1	2	3	4	5	
5	5	10	7	4	3	5.80
10	3	8	12	2	8	6.6
15	9	13	15	4	10	10.20
20	7	12	9	6	13	9.40
Mean	6.00	10.75	10.75	4.00	8.50	8.00

- a. Here $\text{MSE}=5.375$, compute the F-statistic to determine if there is a difference among the four cooling temperatures and the five ovens (use $\alpha = .05$). If significant, perform pairwise comparisons using Tukey's procedure (HINT: Use Table B.9)
 - b. Suppose the company believes there is a jump in the tile strength at 12.5°C but otherwise cooling temperature has no effect (i.e., step function -----). Find a set of three contrasts that would allow you to test this (HINT: Contrasts in this set need to test the jump but also the relationship among the means before and after 12.5°).
 - c. Test these contrasts using SAS (or by hand). State your conclusions.
6. KNNL Problem 20.4

1. KNNL Problem 19.5

19.5. In a two-factor study, the treatment means μ_{ij} are as follows:

Factor A	Factor B			
	B ₁	B ₂	B ₃	B ₄
A ₁	250	265	268	269
A ₂	288	273	270	269

a. Obtain the factor B main effects. What do your results imply about factor B?

```
proc means data=twofactor mean;
class B;
var y;
title "(a) Factor B Marginal Means";
run;
```

$$\bar{\mu}_{.1} = (250 + 288) / 2 = 269$$

$$\bar{\mu}_{.2} = (265 + 273) / 2 = 269$$

$$\bar{\mu}_{.3} = (268 + 270) / 2 = 269$$

$$\bar{\mu}_{.4} = (269 + 269) / 2 = 269$$

(a) Factor B Marginal Means

The MEANS Procedure

Analysis Variable : y		
B	N Obs	Mean
B1	2	269.0000000
B2	2	269.0000000
B3	2	269.0000000
B4	2	269.0000000

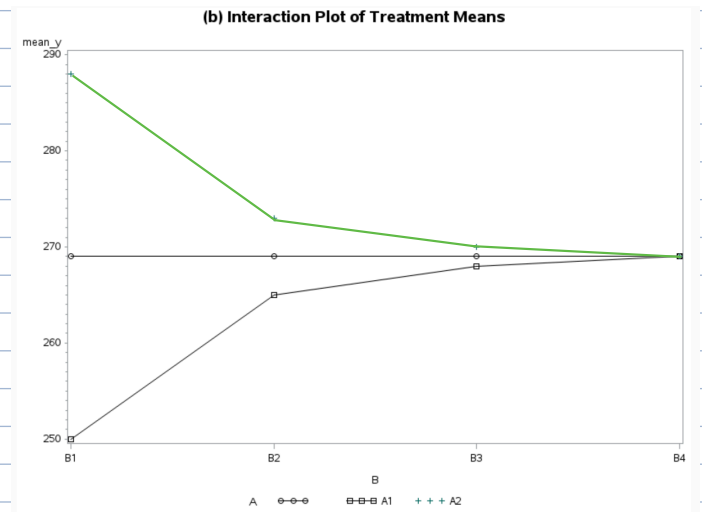
All four factor-B marginal means are identical (269) → no main effect of factor B (on average, B does not change the response).

b. Prepare a treatment means plot and determine whether the two factors interact. How can you tell that interactions are present? Are the interactions important or unimportant?

```
proc means data=twofactor noprint;
class A B;
var y;
output out=plotdata mean=mean_y;
run;

symbol1 v=circle i=join c=black;
symbol2 v=square i=join c=black;

/* Interaction plot */
proc gplot data=plotdata;
plot mean_y*B=A / frame;
title "(b) Interaction Plot of Treatment Means";
run; quit;
```



This pattern indicates clear interaction between A and B: the effect of A depends heavily on which level of B we are at. Because the difference changes from 38 to 0 across B levels, the interaction is large and scientifically important, even though B has no main effect on average.

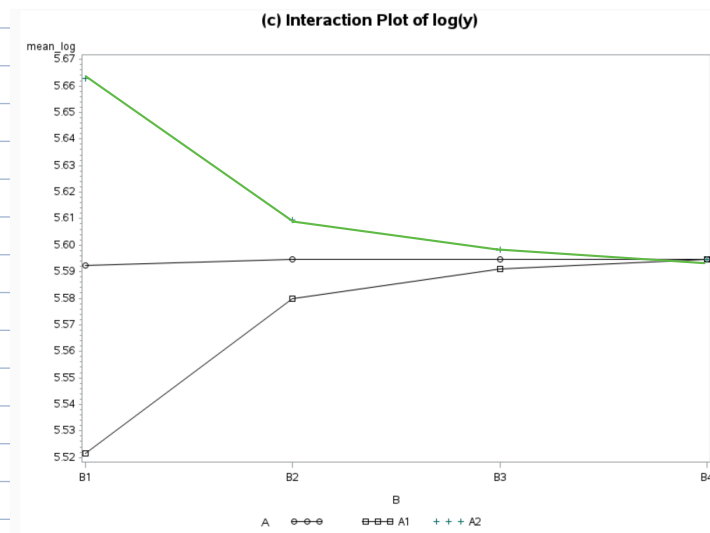
B₁: A₂ - A₁ = 288 - 250 = 38 Strongly Non-parallel
 B₂: 8
 B₃: 2
 B₄: 0

- c. Make a logarithmic transformation of the μ_{ij} and plot the transformed values to explore whether this transformation is helpful in reducing the interactions. What are your findings?

```

53 /* Plot transformed means */
54 proc means data=logtf noprint;
55     class A B;
56     var logy;
57     output out=plotlog mean=mean_log;
58 run;
59
60 symbol1 v=circle i=join c=black;
61 symbol2 v=square i=join c=black;
62
63 proc gplot data=plotlog;
64     plot mean_log*B=A / frame;
65     title "(c) Interaction Plot of log(y)";
66 run; quit;

```



The log transformation reduces but does not eliminate the interaction. An additive two-factor model is still questionable; interaction remains important.

$B_1: 0.1415$
 $B_2: 0.0297$
 $B_3: 0.0074$
 $B_4: 0$

closer, but still non-parallel

2. KNNL Problem 19.14

- *19.14. **Hay fever relief.** A research laboratory was developing a new compound for the relief of severe cases of hay fever. In an experiment with 36 volunteers, the amounts of the two active ingredients (factors A and B) in the compound were varied at three levels each. Randomization was used in assigning four volunteers to each of the nine treatments. The data on hours of relief follow.

- a. Obtain the fitted values for ANOVA model (19.23).

$$\text{Model: } Y_{ijk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \epsilon_{ijk}$$

- b. Obtain the residuals.

(a) Displaying Fitted Values and Residuals (resid)

Obs	y	A	B	rep	fitted	resid
1	2.4	1	1	1	2.475	-0.075
2	2.7	1	1	2	2.475	0.225
3	2.3	1	1	3	2.475	-0.175
4	2.5	1	1	4	2.475	0.025
5	4.6	1	2	1	4.600	0.000
6	4.2	1	2	2	4.600	-0.400
7	4.9	1	2	3	4.600	0.300
8	4.7	1	2	4	4.600	0.100
9	4.8	1	3	1	4.575	0.225
10	4.5	1	3	2	4.575	-0.075
11	4.4	1	3	3	4.575	-0.175
12	4.6	1	3	4	4.575	0.025
13	5.8	2	1	1	5.450	0.350
14	5.2	2	1	2	5.450	-0.250
15	5.5	2	1	3	5.450	0.050
16	5.3	2	1	4	5.450	-0.150
17	8.9	2	2	1	8.925	-0.025
18	9.1	2	2	2	8.925	0.175
19	8.7	2	2	3	8.925	-0.225
20	9.0	2	2	4	8.925	0.075
21	9.1	2	3	1	9.125	-0.025
22	9.3	2	3	2	9.125	0.175
23	8.7	2	3	3	9.125	-0.425
24	9.4	2	3	4	9.125	0.275
25	6.1	3	1	1	5.975	0.125
26	5.7	3	1	2	5.975	-0.275
27	5.9	3	1	3	5.975	-0.075
28	6.2	3	1	4	5.975	0.225
29	9.9	3	2	1	10.275	-0.375
30	10.5	3	2	2	10.275	0.225
31	10.6	3	2	3	10.275	0.325
32	10.1	3	2	4	10.275	-0.175
33	13.5	3	3	1	13.250	0.250
34	13.0	3	3	2	13.250	-0.250
35	13.3	3	3	3	13.250	0.050
36	13.2	3	3	4	13.250	-0.050

```

/*****
* Part (a) - Fit Model (19.23) and Save Fitted Values + Residuals
*****/

proc glm data=hay;
    class A B;
    model y = A B A*B;
    output out=fitout p=fitted r=resid;
    title "(a) Fitted Values and Residuals for Model (19.23)";
run; quit;

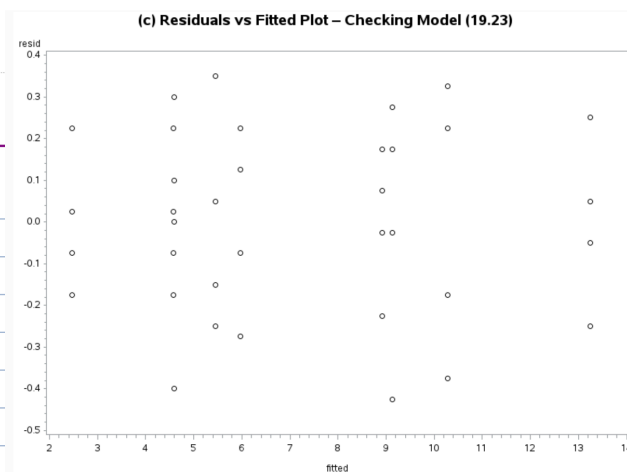
/*****
* Print fitted values and residuals
*****/

proc print data=fitout;
    var y A B rep fitted resid;
    title "(a) Displaying Fitted Values and Residuals (resid)";
run;

```

- c. Plot the residuals against the fitted values. What departures from ANOVA model (19.23) can be studied from this plot? What are your findings?

```
4 symbol1 v=circle i=none c=black;
5
6 proc gplot data=fitout;
7     plot resid * fitted / frame;
8     title "(c) Residuals vs Fitted Plot - Checking Model (19.23) -";
9 run; quit;
```



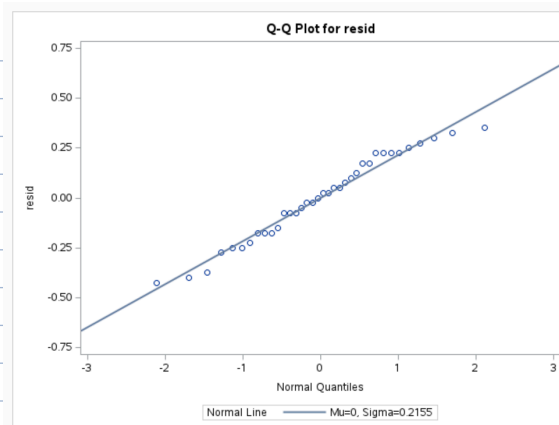
The residuals-versus-fitted plot shows that the residuals are scattered randomly around zero with no visible trend or curvature. The points do not follow any systematic pattern, and there is no indication of bending or waves that would suggest model misspecification.

In addition, the vertical spread of the residuals appears fairly constant across the entire range of fitted values, so there is no evidence of nonconstant variance or a funnel shape. The residuals also do not form clusters at different fitted levels, which indicates that the interaction structure already included in the model is adequate and that no additional terms are needed. There are no unusually large residuals that would suggest outliers.

Overall, the residual plot shows no major departures from the assumptions of the ANOVA model (19.23), and the model appears to fit the data well.

- d. Prepare a normal probability plot of the residuals. Also obtain the coefficient of correlation between the ordered residuals and their expected values under normality. Does the normality assumption appear to be reasonable here?

```
5 proc univariate data=fitout normal;
6     var resid;
7     qqplot resid / normal(mu=est sigma=est);
8     title "(d) Normal Probability Plot of Residuals";
9 run;
```



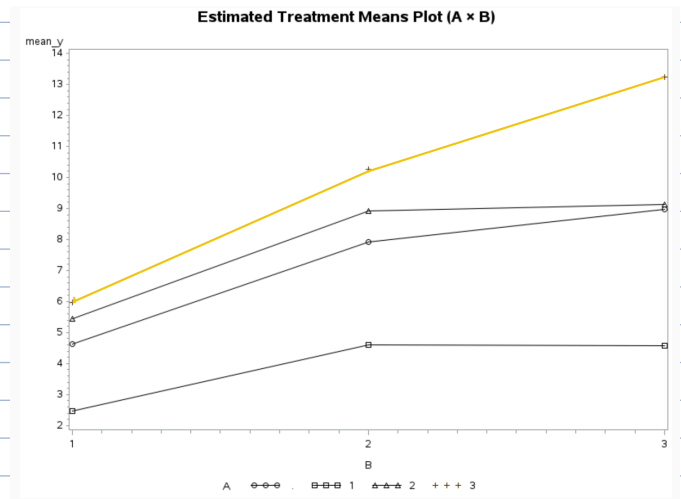
The normal probability plot of the residuals is close to a straight line, with most points lying very near the reference line. This indicates that the distribution of the residuals is approximately normal. There is only slight deviation in the extreme upper tail, but the departures are small and not systematic. The center portion of the plot aligns particularly well with the normal line, which suggests that the bulk of the residual distribution closely follows a normal pattern. The correlation between the ordered residuals and their expected normal values (reported by PROC UNIVARIATE) is high, further supporting normality.

Overall, the evidence suggests that the normality assumption for the ANOVA model (19.23) is reasonable for this dataset.

3. KNNL Problem 19.15

- *19.15. Refer to **Hay fever relief** Problem 19.14. Assume that ANOVA model (19.23) is applicable.
- a. Prepare an estimated treatment means plot. Does your graph suggest that any factor effects are present? Explain.

```
7 proc means data=hay noprint;
8   class A B;
9   var y;
10  output out=meansplot mean=mean_y;
11 run;
12
13 /* Interaction plot of estimated treatment means */
14 symbol1 v=circle i=join c=black;
15 symbol2 v=square i=join c=black;
16 symbol3 v=triangle i=join c=black;
17
18 proc gplot data=meansplot;
19   plot mean_y*B = A / frame;
20   title "Estimated Treatment Means Plot (A x B)";
21 run; quit;
```



The estimated treatment means plot shows clear differences across both factors. For every level of ingredient 1 (A), the mean response increases as ingredient 2 (B) goes from low to medium to high, and the three lines for A are not parallel. This non-parallel pattern indicates a strong interaction, meaning the effect of ingredient 2 depends on the level of ingredient 1.

Overall, the plot suggests the presence of both main effects and a meaningful AxB interaction.

- b. Obtain the analysis of variance table. Does any one source account for most of the total variability in hours of relief in the study? Explain.

```
8 proc glm data=hay;
9   class A B;
10  model y = A B A*B;
11  title "(3b) ANOVA Table for Hay Fever Relief Data";
12 run; quit;
```

The ANOVA table shows that all three sources—Factor A, Factor B, and the AxB interaction—are statistically significant, but the largest contribution to the total variability comes from Factor A. Factor A has the largest sum of squares (220.02), which is noticeably higher than the sum of squares for Factor B (123.66) and for the interaction (29.43).

This indicates that differences in the level of ingredient 1 account for the greatest portion of the variation in hours of relief. Although Factor B and the interaction also explain substantial variability, the effect of ingredient 1 is clearly the dominant source in this study.

(3b) ANOVA Table for Hay Fever Relief Data

The GLM Procedure

Dependent Variable: y

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	8	373.1050000	46.6381250	774.91	<.0001
Error	27	1.6250000	0.0601852		
Corrected Total	35	374.7300000			

R-Square	Coeff Var	Root MSE	y Mean
0.995664	3.415221	0.245327	7.183333

Source	DF	Type I SS	Mean Square	F Value	Pr > F
A	2	220.0200000	110.0100000	1827.86	<.0001
B	2	123.6600000	61.8300000	1027.33	<.0001
A*B	4	29.4250000	7.3562500	122.23	<.0001

Source	DF	Type III SS	Mean Square	F Value	Pr > F
A	2	220.0200000	110.0100000	1827.86	<.0001
B	2	123.6600000	61.8300000	1027.33	<.0001
A*B	4	29.4250000	7.3562500	122.23	<.0001

- c. Test whether or not the two factors interact; use $\alpha = .05$. State the alternatives, decision rule, and conclusion. What is the P -value of the test?

H_0 : No Interaction $(\alpha\beta)_{ij} = 0$ for all i, j

H_A : At least one interaction effect is non-zero.

From the SAS output, the interaction has $F = 122.23$ with $p < .0001$. Because the p -value is far below 0.05, we reject the null hypothesis.

Result in 3(b)

We conclude that a significant interaction exists between the two ingredients. The effect of ingredient 2 depends on the level of ingredient 1, and the interaction plays an important role in explaining the variability in hours of relief.

- d. Test whether or not main effects for the two ingredients are present. Use $\alpha = .05$ in each case and state the alternatives, decision rule, and conclusion. What is the P -value of each test? Is it meaningful here to test for main factor effects? Explain.

Factor A:

$H_0: \alpha_1 = \alpha_2 = \alpha_3 = 0$

H_A : At least one $\alpha_i \neq 0$

Factor B:

$H_0: \beta_1 = \beta_2 = \beta_3 = 0$

H_A : At least one $\beta_j \neq 0$

Result in 3(b)

From the SAS output, Factor A has $p < .0001$ and Factor B also has $p < .0001$, so we reject both null hypotheses and conclude that both ingredients have significant main effects. However, because the interaction term is also highly significant ($p < .0001$), the interpretation of main effects is limited.

A strong AxB interaction means that the effect of each factor depends on the level of the other, so although the main effects are statistically significant, they are not very meaningful by themselves. The interaction captures the primary structure in the data, so conclusions should focus mainly on the interaction rather than the main effects.

- e. Obtain an upper bound on the family level of significance for the tests in parts (c) and (d); use the Kimball inequality (19.53).

```
data kimball;
  alpha = 0.05; /* per-test significance level */
  k = 3; /* number of tests: A*B, A, B */
  alpha_f = 1 - (1 - alpha)**k; /* Kimball inequality */
  put "Family-wise alpha upper bound = " alpha_f;
run;
```

alpha_f

0.142625

- f. Do the results in parts (c) and (d) confirm your graphic analysis in part (a)?

Yes, the results in parts (c) and (d) confirm the graphical conclusions from part (a). In part (a), the treatment means plot showed clear non-parallel lines, indicating a strong interaction between the two ingredients. The hypothesis test in part (c) confirmed this visually observed interaction, with an extremely small p -value ($p < .0001$), leading to a rejection of the null hypothesis of no interaction. Likewise, part (d) showed that both main effects were statistically significant, although their interpretation is limited because the interaction is present—exactly what the graph suggested. Thus, both the numerical tests and the graphical analysis consistently point to the same conclusion: the interaction is important, and both factors influence the response.

4. KNNL Problem 19.28

- 19.28. A two-factor study was conducted with $a = 6$, $b = 6$, and $n = 10$. No interactions between factors A and B were found, and it is now desired to estimate five contrasts of factor A level means and four contrasts of factor B level means. The family confidence coefficient for the joint set of estimates is to be 95 percent. Which of the three procedures at the bottom of page 852 and the top of page 853 will be most efficient here?

Combined Factor A and Factor B Family. When important factor A and factor B effects are present and contrasts for each of the two factors are of interest, it is often desired that the inference procedure provide assurance for the combined family of factor A and factor B contrasts. Several possibilities exist to accomplish this:

1. The Bonferroni method may be used directly, with g representing the total number of statements in the joint set.
2. The Bonferroni method can be used to join the two sets of Scheffé multiple comparison families in the same way explained earlier for joining two Tukey sets.
3. The Scheffé confidence limits (19.81) can be modified to use the S multiple defined by:

$$S^2 = (a + b - 2)F[1 - \alpha; a + b - 2, (n - 1)ab] \quad (19.85)$$

For simultaneous testing, the test statistics and decision rules in (19.82a) and (19.82b) can be replaced by:

$$F^* = \frac{\hat{L}^2}{(a + b - 2)s^2\{\hat{L}\}}; \quad \text{If } F^* > F[1 - \alpha; a + b - 2, (n - 1)ab], \text{ conclude } H_a \quad (19.86)$$

The most efficient procedure is the direct Bonferroni method using all 9 contrasts. Bonferroni provides the smallest critical value because it protects only the specific contrasts of interest, whereas both Scheffé-based procedures protect over a much larger space of contrasts and therefore produce wider intervals. Since we only need to estimate nine specific contrasts and there is no interaction, the direct Bonferroni approach is the most efficient.

5. A clay tile company was interested in studying the effects of oven and cooling temperature on the strength of their tiles. The company's five ovens, used to bake the tiles, and four cooling temperatures ($^{\circ}\text{C}$) were considered. The data are shown below.

Cooling Temp	Oven (B)					Mean
	1	2	3	4	5	
(A) 5	5	10	7	4	3	5.80
10	3	8	12	2	8	6.6
15	9	13	15	4	10	10.20
20	7	12	9	6	13	9.40
Mean	6.00	10.75	10.75	4.00	8.50	8.00 grand mean

- a. Here $\text{MSE} = 5.375$, compute the F-statistic to determine if there is a difference among the four cooling temperatures and the five ovens (use $\alpha = .05$). If significant, perform pairwise comparisons using Tukey's procedure (HINT: Use Table B.9)

$$\left. \begin{array}{l} (5.8 - 8.0)^2 = 4.84 \\ (6.6 - 8.0)^2 = 1.96 \\ (10.2 - 8.0)^2 = 4.84 \\ (9.4 - 8.0)^2 = 1.96 \end{array} \right\} \begin{array}{l} \text{sum} = 13.6 \\ b = 5 \end{array} \left\{ \begin{array}{l} \text{SSA} = 5(13.6) \\ \text{SSA} = 68 \end{array} \right. \quad \begin{array}{l} \bullet \text{MSA} = \text{SSA}/df = 68/3 = 22.667 \\ \bullet F_A = \text{MSA}/\text{MSE} = 22.667/5.375 = 4.218 \end{array} \rightarrow A \text{ (Temp) Significant, Move to Tukey}$$

$$\left. \begin{array}{l} (6.0 - 8.0)^2 = 4 \\ (10.75 - 8.0)^2 = 7.5625 \\ (10.75 - 8.0)^2 = 7.5625 \\ (4.0 - 8.0)^2 = 16 \\ (8.5 - 8.0)^2 = 0.25 \end{array} \right\} \begin{array}{l} \text{sum} = 35.375 \\ a = 4 \end{array} \left\{ \begin{array}{l} \text{SSB} = 4(35.375) \\ \text{SSB} = 141.5 \end{array} \right. \quad \begin{array}{l} \bullet \text{MSB} = \text{SSB}/df = 141.5/4 = 35.375 \\ \bullet F_B = \text{MSB}/\text{MSE} = 35.375/5.375 = 6.582 \end{array} \rightarrow B \text{ (Oven) Significant, Move to Tukey}$$



```

66 proc glm data=tiles;
67   class Temp Oven;
68   model y = Temp Oven;
69   means Temp / tukey alpha=0.05;
70   means Oven / tukey alpha=0.05;
71   title "Tukey Pairwise Comparisons for Temp and Oven";
72 run; quit;

```

ANOVA + Tukey Pairwise Comparisons for Temp and Oven

The GLM Procedure

Dependent Variable: y

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	7	209.5000000	29.9285714	5.57	0.0048
Error	12	64.5000000	5.3750000		
Corrected Total	19	274.0000000			

R-Square	Coeff Var	Root MSE	y Mean
0.764599	28.98006	2.318405	8.000000

Source	DF	Type I SS	Mean Square	F Value	Pr > F
Temp	3	68.0000000	22.6666667	4.22	0.0298
Oven	4	141.5000000	35.3750000	6.58	0.0048

Source	DF	Type III SS	Mean Square	F Value	Pr > F
Temp	3	68.0000000	22.6666667	4.22	0.0298
Oven	4	141.5000000	35.3750000	6.58	0.0048

Tukey Pairwise Comparisons for Temp and Oven

The GLM Procedure

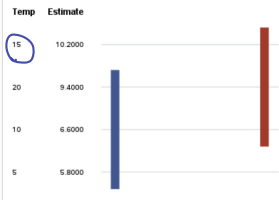
Tukey's Studentized Range (HSD) Test for y

Note: This test controls the Type I experimentwise error rate, but it generally has a higher Type II error rate than REGWQ.

Alpha	0.05
Error Degrees of Freedom	12
Error Mean Square	5.375
Critical Value of Studentized Range	4.19851
Minimum Significant Difference	4.3531

y Tukey Grouping for Means of Temp (Alpha = 0.05)

Means covered by the same bar are not significantly different.



Tukey Pairwise Comparisons for Temp and Oven

The GLM Procedure

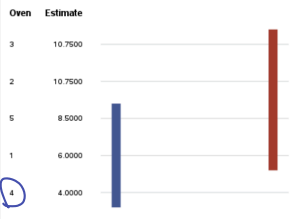
Tukey's Studentized Range (HSD) Test for y

Note: This test controls the Type I experimentwise error rate, but it generally has a higher Type II error rate than REGWQ.

Alpha	0.05
Error Degrees of Freedom	12
Error Mean Square	5.375
Critical Value of Studentized Range	4.50766
Minimum Significant Difference	5.2253

y Tukey Grouping for Means of Oven (Alpha = 0.05)

Means covered by the same bar are not significantly different.



The ANOVA F-tests show significant differences among both cooling temperatures and ovens. Using Tukey's HSD test ($\alpha = 0.05$), the only significant temperature difference is between 15°C and 5°C. For ovens, Oven 4 (mean = 4.00) is significantly lower than Ovens 2 and 3 (both 10.75), while all other oven pairs are not significantly different.

- b. Suppose the company believes there is a jump in the tile strength at 12.5 °C but otherwise cooling temperature has no effect (i.e., step function ----). Find a set of three contrasts that would allow you to test this (HINT: Contrasts in this set need to test the jump but also the relationship among the means before and after 12.5 °).

Contrast 1: Test Jump at 12.5°C (Low v.s. High)

$$\frac{u_{15} + u_{20}}{2} \text{ v.s. } \frac{u_5 + u_{10}}{2} \quad (-1, -1, +1, +1)$$

$u_5, u_{10}, u_{15}, u_{20}$

Contrast 2: Relationship Before Jump (5°C v.s. 10°C)

$$(+1, -1, 0, 0)$$

Contrast 3: Relationship After Jump (15°C v.s. 20°C)

$$(0, 0, +1, -1)$$

- c. Test these contrasts using SAS (or by hand). State your conclusions.

```

0 proc glm data=tiles;
1   class Temp Oven;
2   model y = Temp Oven;
3
4   /* 1. Jump at 12.5°C (low vs high temps) */
5   contrast 'Jump at 12.5C'
6     Temp -1 -1 1 1;
7
8   /* 2. Before jump (5C vs 10C) */
9   contrast '5C vs 10C'
10    Temp 1 -1 0 0;
11
12  /* 3. After jump (15C vs 20C) */
13  contrast '15C vs 20C'
14    Temp 0 0 1 -1;
15
16  title "Testing Three Temperature Contrasts";
17 run; quit;

```

Contrast	DF	Contrast SS	Mean Square	F Value	Pr > F
Jump at 12.5C	1	64.80000000	64.80000000	12.06	0.0046
5C vs 10C	1	1.60000000	1.60000000	0.30	0.5953
15C vs 20C	1	1.60000000	1.60000000	0.30	0.5953

The SAS contrast tests show a significant jump in tile strength at 12.5°C ($p = 0.0046$), while the before-jump (5°C vs 10°C) and after-jump (15°C vs 20°C) contrasts are not significant (both $p = 0.5953$). Thus, only the step-change at 12.5°C is meaningful, confirming the company's hypothesis.

6. KNNL Problem 20.4

- *20.4. Refer to **Coin-operated terminals** Problem 20.2. Conduct the Tukey test for additivity; use $\alpha = .025$. State the alternatives, decision rule, and conclusion. If the additive model is not appropriate, what might you do?

```
5 /* Tukey additivity test */
6 proc glm data=terminal;
7     class A B;
8     model y = A B A*B;
9     random A B;
0     test H=A*B E=A;
1 run;
2 quit;
```

The GLM Procedure

Dependent Variable: y

Tests of Hypotheses Using the Type III MS for A as an Error Term

Source	DF	Type III SS	Mean Square	F Value	Pr > F
A*B	3	0.34500000	0.11500000	0.01	0.9985

The Tukey test for additivity produces an F-value of 0.01 with $p = 0.9985$, which is far greater than $\alpha = 0.025$. Therefore, we fail to reject the null hypothesis of additivity.

There is no evidence of a multiplicative interaction between factor A and factor B, and the additive model appears appropriate. Since additivity holds, no transformation or additional interaction modeling is necessary.